ABSTRACT

ForgeOS is a modular Operating Systems project developed to explore and demonstrate the practical concepts of operating systems and Linux-based system programming. The project brings together different components under a unified project structure, focusing on the interaction between user-level applications and operating system functionalities. It is organized into two major modules: NanoKernel and ShellForge, each representing a distinct part of the project.

The NanoKernel module contains QueueLess, while the ShellForge module includes LinuxCore, a custom command-line shell developed using the C programming language in a Linux environment. LinuxCore provides an interactive terminal interface that allows users to execute built-in commands and external Linux programs. It incorporates functionalities such as directory navigation, command history, system information retrieval, CPU and memory monitoring, process listing, disk usage information, file creation, file reading and writing, and file permission inspection. Through these functionalities, LinuxCore demonstrates important Operating Systems concepts, including process management, file handling, command interpretation, and interaction with Linux system interfaces.

The development of ForgeOS provides practical exposure to modular software development, Linux environments, C programming, and fundamental operating system concepts. It bridges the gap between theoretical knowledge and practical implementation by organizing different components into a single project. Overall, ForgeOS serves as an educational platform for understanding operating system functionalities, system-level programming, and the design of modular computing systems.